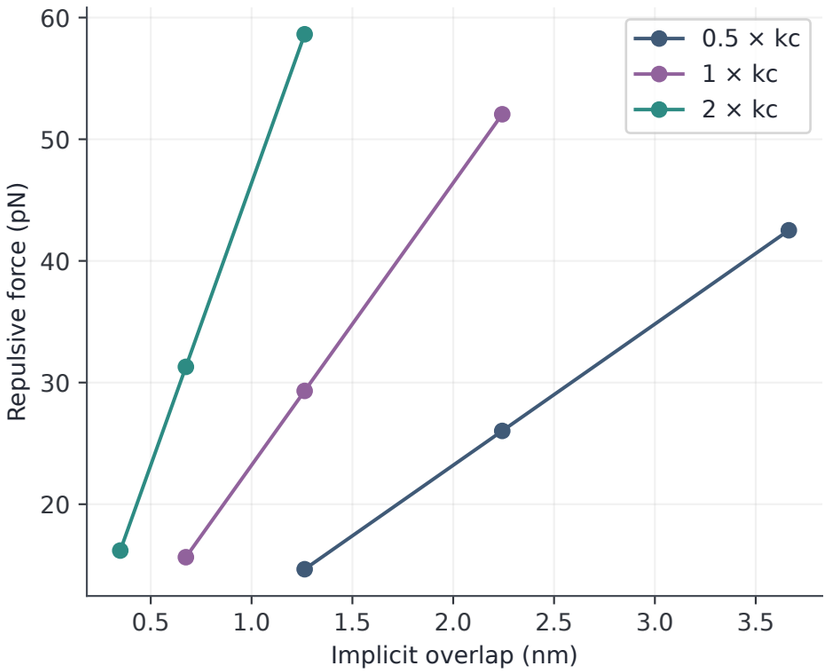
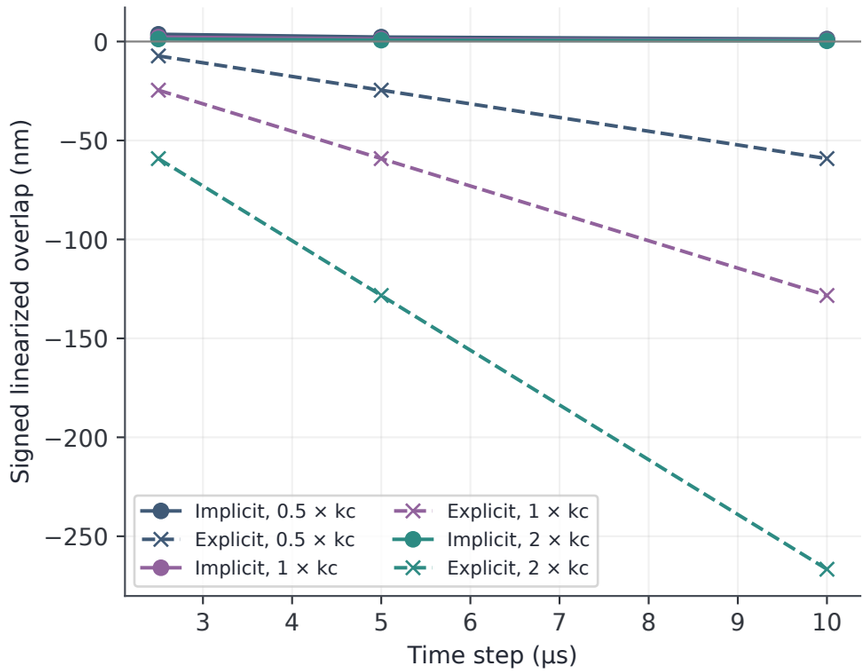
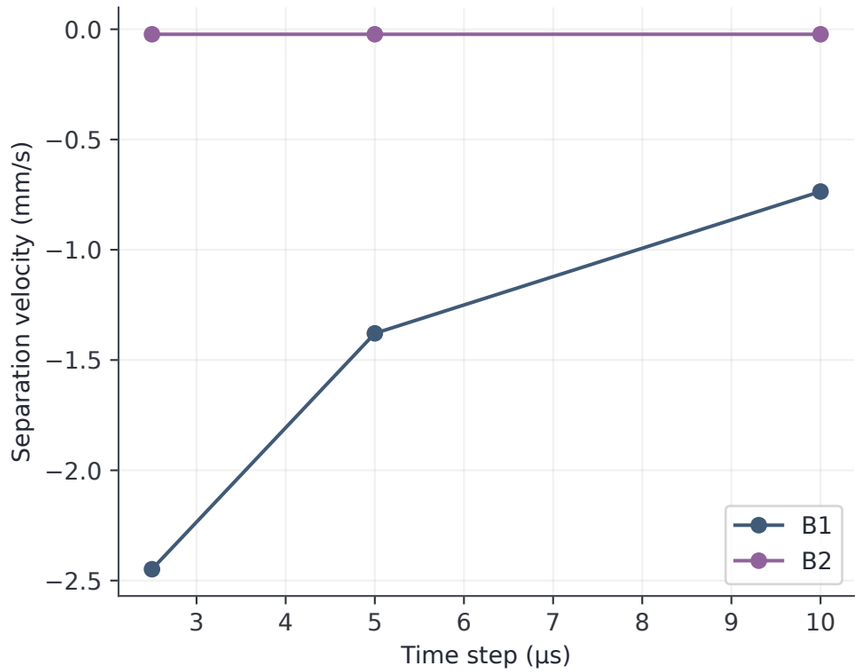
a Candidate spring: force and overlap



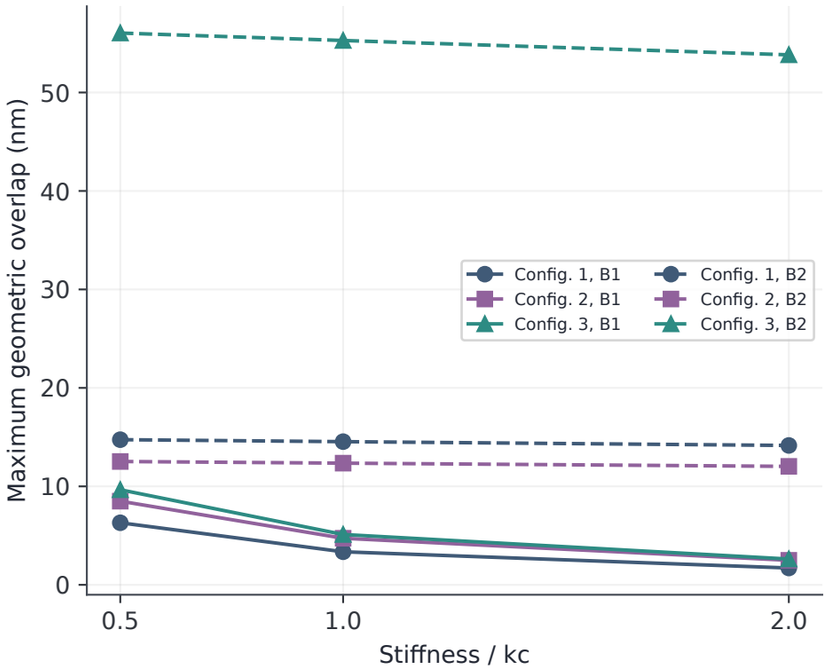
b Explicit overshoot versus implicit decay



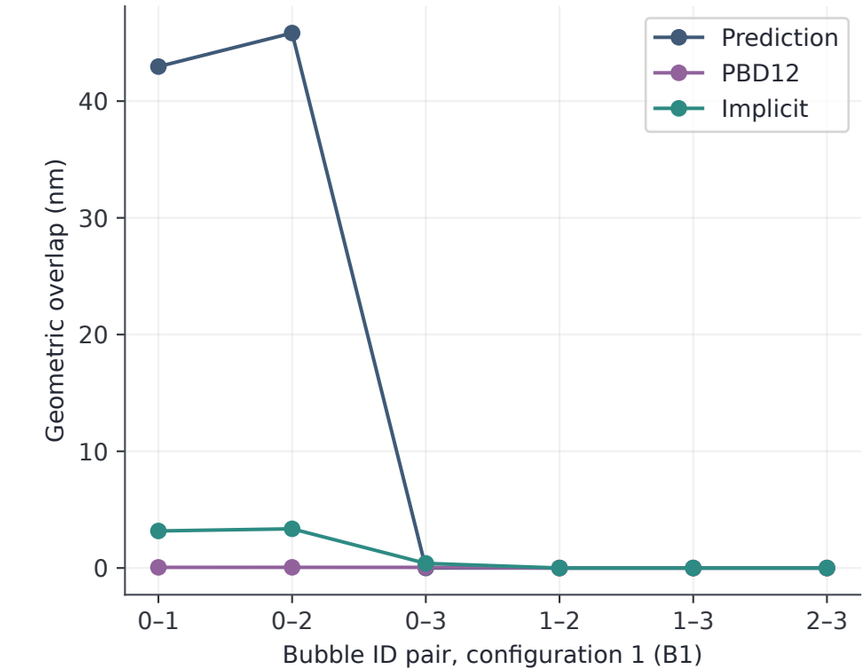
c Normal velocity under compression



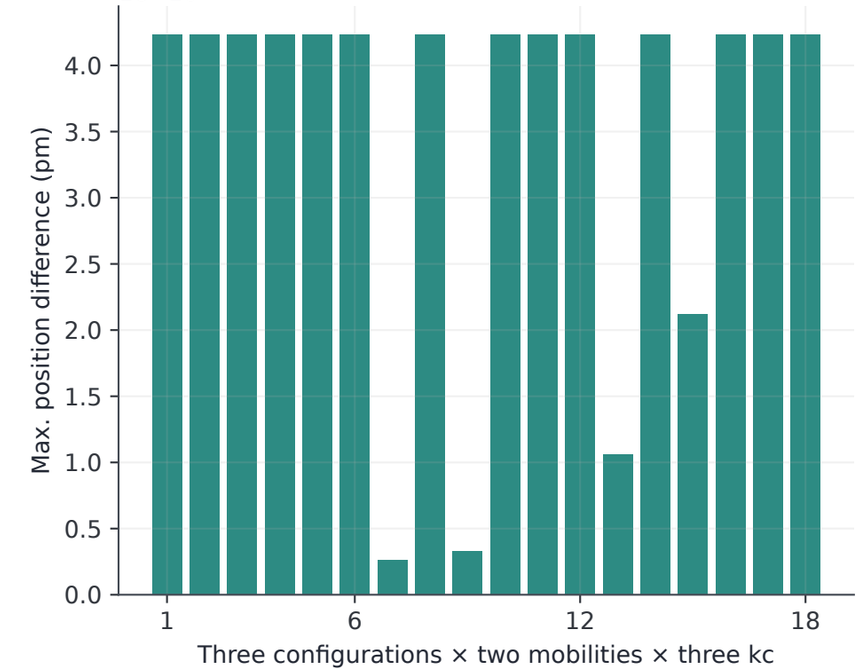
d Four bubbles: stiffness sensitivity



e Shared-node contact redistribution



f Independent four-bubble force reference



kc = 0.0232066 N/m, selected from saved compression forces / 8.9 nm before contact trials; not a measured SonoVue stiffness.
90 two-bubble cases plus ID swaps and hard-limit check; 18 four-bubble cases. B1: free Stokes; B2: inverse of the original lubrication resistance.